Example 9.14

Compute $\lim_{x\to 1}\dfrac{\sqrt{x}-1}{x-1}$.

Solution

Here $\lim_{x\to 1}(x-1)=0$. In such cases, rationalise the numerator.

$$\lim_{x\to 1}\frac{\sqrt{x}-1}{x-1}=\lim_{x\to 1}\frac{(\sqrt{x}-1)}{(\sqrt{x}-1)(\sqrt{x}+1)}=\lim_{x\to 1}\frac{1}{\sqrt{x}+1}=\frac{\lim_{x\to 1}(1)}{\lim_{x\to 1}\left(\sqrt{x}+1\right)}=\frac{1}{2}.$$

Example 9.15

Find $\lim_{t\to 0}\dfrac{\sqrt{t^2+9}-3}{t^2}$.

Solution

We can't apply the quotient theorem immediately. Use the algebra technique of rationalising the numerator.

$$\frac{\sqrt{t^2+9}-3}{t^2}=\frac{\left(\sqrt{t^2+9}-3\right)\left(\sqrt{t^2+9}+3\right)}{t^2\left(\sqrt{t^2+9}+3\right)}=\frac{t^2+9-9}{t^2\left(\sqrt{t^2+9}+3\right)}$$

$$\lim_{t\to 0}\frac{\sqrt{t^2+9}-3}{t^2}=\lim_{t\to 0}\frac{t^2}{t^2\sqrt{t^2+9}+3}=\lim_{t\to 0}\frac{1}{\sqrt{t^2+9}+3}=\frac{1}{\sqrt{9}+3}=\frac{1}{6}.$$

Theorem 9.4

$$\lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}.$$

Proof

We know that $x^n-a^n=(x-a)(x^{n-1}+x^{n-2}a+x^{n-3}a^2+\cdots+xa^{n-2}+a^{n-1})$

$$\lim_{x\to a}\frac{x^n-a^n}{x-a}=\lim_{x\to a}\frac{(x-a)(x^{n-1}+x^{n-2}a+x^{n-3}a^2+\cdots+xa^{n-2}+a^{n-1})}{(x-a)}$$

$$=\lim_{x\to a}(x^{n-1}+x^{n-2}a+x^{n-3}a^2+\cdots+xa^{n-2}+a^{n-1})$$

$$=a^{n-1}+a^{n-1}+\cdots+a^{n-1}\ (n\text{ times})$$

$$\lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}\ .$$

It is also true for any rational number n. ∎

Example 9.16

Compute $\lim_{x\to 1}\dfrac{x^3-1}{x-1}$.

Solution

$$\lim_{x\to 1}\frac{x^3-1}{x-1}=\lim_{x\to 1}\frac{x^3-1^3}{x-1}=3(1)^{3-1}=3\ .$$

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